

# Sudip Pan

## Electronics and Communication Engineering

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Portfolio: [Sudip's Portfolio](#)

### Summary

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Electronics and Communication Engineering student with experience in C++, Python, analog CMOS design, and desktop application development. Seeking a Graduate Engineer Trainee role to apply software and electronics engineering skills.

### Skills

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**Languages:** C • C++ • Python • JavaScript • HTML/CSS

**Frameworks:** React • Vite • Tailwind CSS • OpenGL • GLFW • Dear ImGui

**Tools:** Cadence Virtuoso • Git • GitHub • CMake • Rider • Visual Studio • VS Code • MS Office

**Core Skills:** Circuit Design • PCB Design • Desktop Application Development • Analog IC Design

### Projects

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#### 1. CMOS Instrumentation Amplifier Design using Cadence Virtuoso

[portfolio-link](#)

Designed and implemented a two-stage CMOS operational amplifier using 90 nm CMOS technology in Cadence Virtuoso. Building upon the validated op-amp design, developed a CMOS instrumentation amplifier optimized for analog signal conditioning applications.

- Designed and simulated a two-stage CMOS operational amplifier using 90 nm technology in Cadence Virtuoso.
- Developed a CMOS instrumentation amplifier and optimized gain, bandwidth, stability, and power.
- Research paper accepted for presentation at **ICICET 2026** (Pending publication).

#### 2. PDF Graph Digitizer & Bode Plot Analyzer | Python

[github-link](#)

Developed a Python-based desktop application to digitize and replot CMOS amplifier simulation graphs from PDF reports, addressing the lack of suitable graph extraction tools during analog IC design research.

- Built a Python desktop application to digitize and replot CMOS simulation graphs from PDF reports.
- Implemented interactive graph extraction, Bode plot analysis, and CSV export.
- Developed to simplify Cadence Virtuoso simulation analysis.

#### 3. code ZEN – Lightweight Code Editor & Developer Tool | C++, OpenGL, GLFW, Dear ImGui

[github-link](#)

Designed and developed a lightweight code editor for Windows after identifying the high resource usage of traditional IDEs while developing Unreal Engine projects. Focused on delivering a responsive development environment with essential coding and terminal features.

- Developed a lightweight C++ code editor using OpenGL, GLFW, Dear ImGui, and Win32 ConPTY.
- Implemented docking UI, integrated terminal, asynchronous file indexing, and compile/run workflows.
- Structured the application using CMake and multithreading.

#### 4. Personal Portfolio Website | React.js, Vite, Tailwind CSS

[portfolio-link](#)

Designed and developed a responsive personal portfolio website to showcase software development, game development, and electronics engineering projects.

- Built a responsive portfolio using React.js, Vite, and Tailwind CSS.
- Showcases software, electronics, and game development projects.

Experience

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Eastern Railway S &T Department

Kavach System: Safety Integration

- Studied the **Kavach** Automatic Train Protection (ATP) system and its role in enhancing railway safety and preventing collisions.
- Gained an understanding of railway signaling, train communication, and safety integration concepts used in modern railway operations.
- Observed the implementation and operation of signaling and telecommunication equipment in a safety-critical railway environment.

Education

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Ramkrishna Mahato Government Engineering College, Purulia

2022 – 2026

Bachelor of Technology — Electronics and Communication Engineering

Class XII

WBCHSE --- Ramjibanpur Babulal Institution (91.6%)

2021

Class X

WBBSE --- Ramjibanpur Babulal Institution (93.57%)

2019